



Motherboards

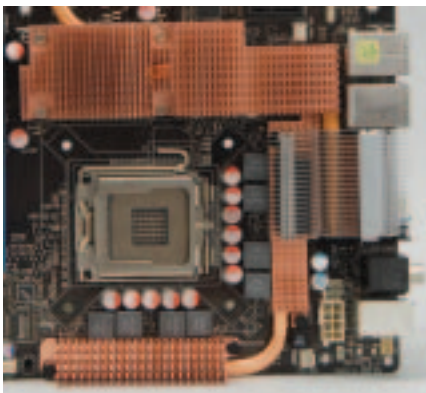
When choosing a motherboard, the most important thing to consider is its price. It's all too easy to fall prey to a bloated list of features you don't need. Boards these days are moving away from the mundane platforms they used to be and some high-end boards have extra features like fancy lighting, heat-pipe cooling solutions among other nifty add-ons. Some of these features may or may not appeal, while some may be useful to you and some may be a total waste of cash.

Video Connectivity



If your motherboard has an onboard graphics solution, it will definitely have some sort of video connectivity. D-Sub or DVI are the usual connects that most motherboards have and most boards will usually have either of the two connects. HDMI is a new standard that allows digital audio and video on a single connect. Some of the newer boards have an HDMI port, which is a plus for anyone in general and HTPC users in particular.

Cooling

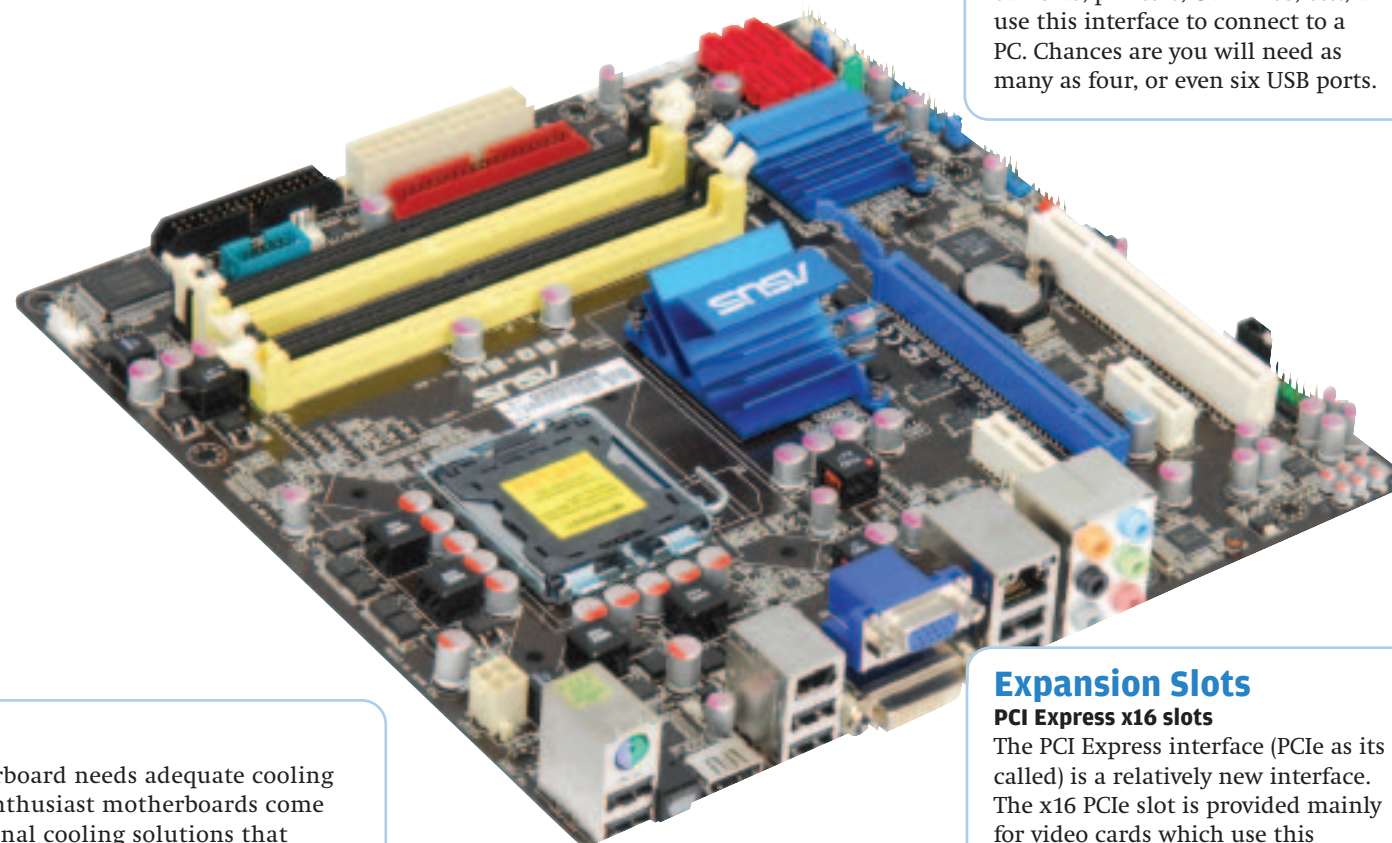


Your motherboard needs adequate cooling too. Some enthusiast motherboards come with additional cooling solutions that utilise heat-pipes to whisk heat away from the hot regions. These may be worth checking out as lower temperatures generally increase both the longevity and stability of your motherboard. Although this feature may not be essential to everyone, if you are building a powerful computer it may be worth looking at.

Addons



Enthusiast motherboards come with some rather unique features. Some of them may have power on / off, reset and clear CMOS switches on the board PCB or the rear I/O panel. These are more than flashy add-ons and of great use to tweekers and overclockers.



Memory Slots

DDR RAM is history these days and its hard to find any in the market. Your motherboard should ideally support DDR2 memory with four memory slots on the board. This gives you the flexibility of opting for lower density, cheaper DIMMs or higher density costlier DIMMs. Some motherboards support both DDR2 and DDR3 — an added plus because DDR3 will become the *de facto* standard within a year and your motherboard will no become obsolete as other DDR2-only boards will be.



USB Ports



USB ports are necessary for external connectivity and are usually found on the rear panel of your motherboard. Digital cameras, printers, USB mice, etc., all use this interface to connect to a PC. Chances are you will need as many as four, or even six USB ports.

SATA Ports

Serial ATA is the existing storage interface for hard drives and motherboards these days. The number of



ports you need depends on your storage needs and the number of hard drives you have. It's a good practice in future proofing to look for a motherboard with at least four SATA ports.

Audio

Most motherboards come with either 5.1-channel or 7.1-channel audio connectivity. This is via analog 3.5-mm connects. This is important for every user as your speakers would connect here. A microphone connect is also provided. In addition some boards support digital connects like Co-axial (SPDIF) or optical connectivity. Look out for this as digital connectivity usually means better signal quality and less interference.



Audio Connectors
Analog interface

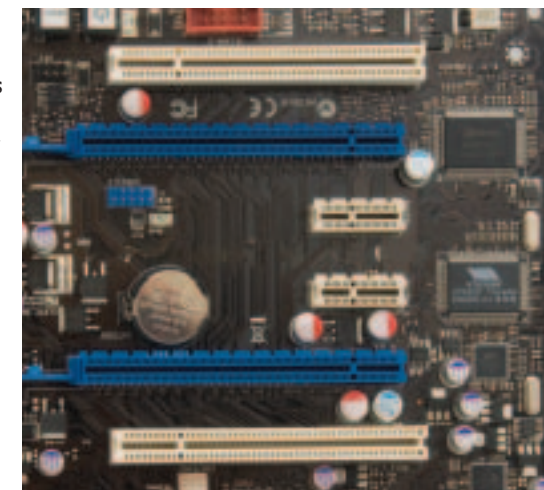


SPDIF/Optical
Digital connectivity

Expansion Slots

PCI Express x16 slots

The PCI Express interface (PCIe as it is called) is a relatively new interface. The x16 PCIe slot is provided mainly for video cards which use this interface as it provides very high bandwidth. A PCI Express slot is of great importance to anyone interested in using a graphics card. Gamers and extreme enthusiasts may want to look at two such slots for using multiple video cards and enabling multi-GPU support technologies like NVIDIA's SLI (Scalable Linked Interface) or ATI's Crossfire.



PCI slots

These are still used in some video mixing cards. PCI slots are typically white. Some motherboards have the PCI slots adjacent to the graphics cards x16 slot, making it unusable.

PCI Express x1/x4 slot

The PCIe X1 slot is hardly in use except by the newer sound cards. However, an X1 slot certainly cannot hurt.